

ANDRÉ ROGÉRIO DA SILVA FILHO

Software Engineer | Applied AI · Enterprise Systems · Cloud

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SUMMARY

Software Engineer building at the intersection of applied AI, enterprise systems, and cloud infrastructure. Creator of an AI-powered ITSM platform piloted with 22 internal users, an industrial VR training simulator defended as a capstone project at a technology residency, and a cognitive memory research platform evolving into an academic paper. Core stack: TypeScript, React, Node.js, PostgreSQL, Google Gemini API, and Microsoft Graph.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Node.js, React 18/19, Next.js, Express, Fastify, Three.js, React Three Fiber, TailwindCSS

Applied AI: Google Gemini API, Structured Outputs, JSON Schemas, Vector Embeddings, Semantic Search, RAG Patterns, LLM Integration

Data & Infrastructure: PostgreSQL, Redis, Supabase, Firebase, Prisma ORM, Docker, REST APIs, WebSockets, Git

Enterprise & Cloud: Google Cloud, Microsoft Graph API, Entra ID, Intune, Microsoft 365, SAP, AES-256-GCM Encryption, LGPD/GDPR Compliance

PROJECTS

Memory Lens | Cognitive Computing & Spatial Interface

[GitHub](#)

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Firebase, Gemini API, Vitest

- Designed and built a cognitive memory platform modeling human memory retention, associative recall, and structured forgetting using vector embeddings, episodic clustering, and Ebbinghaus-based exponential decay, reducing token storage by up to 80%.
- Engineered a 3D spatial interface ("Chief Organism") in React Three Fiber with custom GLSL shaders synchronized to system state via an event bus.
- Built 16 REST API endpoints with Gemini structured JSON schemas and IDOR security test coverage in Vitest. Evolved into formal academic research under Prof. Dr. Fabricio Londero at UniFacens and is now being developed into an academic paper.

FundiVR | Industrial Digital Twin & VR Training Simulator

[GitHub](#)

React 18, TypeScript, Fastify, PostgreSQL, Prisma ORM, WebSockets, Docker, Vitest

- Architected a full-stack digital twin simulator for training aluminum furnace operators in 700°C+ environments, with operational state machines covering PPE checks, thermal control, skimming, and emergency shutdown procedures.
- Engineered real-time bi-directional telemetry over WebSockets in a modular pnpm monorepo, with an automated scoring engine evaluating precision, reaction time, and SOP compliance.
- Built a decoupled zero-dependency demo mode enabling standalone operation without database connectivity.

Enterprise ITSM & AI Assistant

[GitHub](#)

Next.js 16, TypeScript, Supabase, PostgreSQL, Gemini 2.5 Flash, Microsoft Graph API, Redis, Zod

- Built an ITSM platform with intelligent ticket deflection, automated triage, and SLA tracking, piloted with 22 internal users.
- Implemented a Confined AI pattern: LLM prompt → typed JSON output → Zod schema validation → fallback business rules → application action.
- Integrated Microsoft Graph API for real-time Intune device health and compliance telemetry during ticket intake.

Onboarding OS | Multi-Tenant SaaS

[GitHub](#)

React 19, TypeScript, Node.js, Express, PostgreSQL, Supabase, Redis, Zod

- Designed a multi-tenant platform automating employee onboarding across HR, IT, and managers, with PostgreSQL Row-Level Security for tenant isolation and AES-256-GCM document encryption.
- Built scheduled background workers for LGPD data lifecycle compliance, automatically anonymizing sensitive records upon policy expiration.

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura)
IT Analyst | São Paulo, Brazil

06/2026 - Present

- Develop internal automation and applied AI systems, including "Isa", a business assistant powered by Google Gemini with structured outputs and deterministic validation.
- Implement enterprise integrations with Microsoft Graph API, Entra ID, and Intune for fleet compliance monitoring and identity lifecycle automation.
- Manage ERP integrations and data integrity across operational and financial workflows.

Farmamed Drogarias
Support & Infrastructure Analyst | Sorocaba, Brazil

10/2025 - 03/2026

- Administered multi-site endpoint and identity infrastructure across Microsoft 365, Entra ID, Active Directory, and Intune.
- Developed Python automation scripts for operational task streamlining and log analysis.
- Handled N2 support escalations under strict SLA requirements.

Wobben Windpower
Systems & IT Support Intern | Sorocaba, Brazil

11/2023 - 04/2025

- Selected for the global Microsoft Copilot Flight Crew program, evaluating generative AI enterprise features and adoption workflows.
- Supported enterprise service desk operations via ServiceNow with SLA-driven incident resolution.
- Administered user access and permissions across SAP ERP, Active Directory, and Microsoft 365.

EDUCATION

Higher Technology Degree in Systems Analysis and Development
Centro Universitário SENAC, Brazil

2026

RESEARCH & TECHNOLOGY RESIDENCY

Residency in Digital Twins & 5G Systems
UniFacens | Softex | Ministry of Science, Technology and Innovation (MCTI)

2025 - 2026

- Selected through a competitive national process for a 360-hour intensive program covering AI/ML, Industrial IoT, XR, and 5G architecture.
- Technical capstone: FundiVR, an industrial digital twin and VR training simulator defended before an academic and industry panel.

Academic Research | Cognitive Computing & Spatial Interfaces
UniFacens | Advisor: Prof. Dr. Fabricio Londero

2025 - Present

- Researching semantic document clustering, cognitive forgetting models, and 3D spatial interfaces through the Memory Lens platform.
- Research is currently being developed into an academic paper.

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader | Certified AppSec Practitioner (CAP) | CCNA 1: Introduction to Networks (Cisco) | Network Security & Endpoint Security (Cisco)

Languages: English (C1+) | Portuguese (Native) | Spanish (Elementary)